EcoGen-AI — Technical Milestone Report

Layer 1 (Ingestion & Monitoring) Finalization, Metrics Expansion & Hardening

Target System: EcoGen-AI Ingestion Layer	**Database:** SQLite (ecogen.db)
Stack: FastAPI, SQLAlchemy, Pydantic, SQLite	**Migration Status:** Completed & Non-Destructive
Verification Result: 6 / 6 Test Suites Passed	**Deployment Readiness:** Layer 2 Ready

1. Executive Summary

This document confirms the completion and end-to-end verification of **Layer 1 (Ingestion & Monitoring)** for the **EcoGen-AI** platform. EcoGen-AI operates across three integrated layers: (1) Ingestion & Monitoring, (2) Analytical Engine (Energy & Carbon computation), and (3) Strategy & Optimization. To enable Layer 2 to calculate accurate kilowatt-hours, embodied carbon, and marginal emissions, Layer 1 was hardened with missing operational metadata (datacenter PUE, GPU specifications, MoE active parameters, idle baseline power, and hardware lifespans). All modifications were executed non-destructively, preserving existing database rows and maintaining backward compatibility.

2. System Architecture & Metric Flow

Layer	Role	Inputs from Layer 1
Layer 1 Ingestion	Intercepts LLM inference calls and hardware states; logs runtime, tokens, power draw, and datacenter parameters.	Ingests: InferenceEvent & HardwareSnapshot
Layer 2 Analytics	Computes dynamic energy (kWh), marginal carbon footprint (gCO2eq), incremental power over baseline, and embodied emissions amortization.	Requires: pue, num_gpus_used, gpu_model, active_param_count_b, idle_power_baseline_w
Layer 3 Strategy	Recommends model routing, batch size tuning, region carbon-intensity shifting, and MoE vs dense optimizations.	Requires: provider, batch_size, department, cost_usd, use_case

3. Schema & Metric Additions

3.1 InferenceEvent Table Extensions

Field Name	SQL Type	Default / Null	Purpose & Role in Layer 2 / 3
provider	VARCHAR	NULL	Origin provider (e.g., 'gemini', 'ollama') for cross-provider comparative analytics.
pue	FLOAT	1.2	Power Usage Effectiveness for cooling/datacenter multiplier in total energy calculation.
gpu_model	VARCHAR	NULL	Accelerator hardware spec (e.g., 'H100', 'A100-SXM4-80GB', 'none') for TDP lookups.
num_gpus_used	INTEGER	1	Scale multiplier for multi-GPU distributed model parallelism.
batch_size	INTEGER	1	Batch throughput optimization tracking for Layer 3 recommendation engine.
cost_usd	FLOAT	NULL	Financial cost per request for carbon-cost pareto efficiency frontier analysis.
department	VARCHAR	NULL	Internal cost center / departmental ESG attribution reporting.
use_case	VARCHAR	NULL	Workload categorization (e.g., 'code-refactoring', 'customer-chat', 'summarization').
param_count_b	FLOAT	NULL	Total model parameter count (Billions). Basis for Tier 2 FLOP/token energy estimation.
active_param_count_b	FLOAT	NULL	Active parameter count for MoE models (e.g. 12.9B for Mixtral 8x7B) vs total parameters.

3.2 HardwareSnapshot Table Extensions

Field Name	SQL Type	Default / Null	Purpose & Role in Layer 2 / 3
idle_power_baseline_w	FLOAT	NULL	Resting baseline power. Enables Layer 2 to isolate incremental workload energy: P_incremental = P_active - P_idle.
hardware_install_date	DATETIME	NULL	Deployment timestamp for hardware manufacturing embodied carbon amortization over expected lifespan (GHG Scope 3).

4. Hardening & Collector Upgrades

4.1 LLM Metadata Collector (collectors/gen_ai_data.py & genai_metadata.py)
• **Static Parameter Mapping (MODEL_PARAM_MAP):** Best-effort static table pre-populates parameter counts for Gemma (270m to 27b), LLaMA (1b to 405b), Mistral/Mixtral (7b, 8x7b, 8x22b), Gemini (1.5/2.0/2.5/3.5 Flash & Pro), Qwen 2.5, and Phi.
• **MoE Active Parameter Resolution (MOE_ACTIVE_PARAM_MAP):** Differentiates between total and active routed parameters (e.g., Mixtral 8x7B has 46.7B total but only 12.9B active per token). Dense models automatically default active = total.
• **Provider Persistence:** provider parameter in @track_inference is now posted into /ingest/event and stored in SQLite.
• **Token Extraction Resilience:** Extended _extract_tokens to handle both object attributes (usage_metadata) and dictionary responses.

4.2 Hardware Power Collector (collectors/hardware_data.py & hardware_power.py)
• **Baseline Power Sampling:** Added get_idle_power_baseline(samples=4, interval_sec=0.5) to measure resting load.
• **API Integration:** Exposed optional automatic sampling via POST /ingest/hardware?sample_idle=true.
• **Helper Function:** Added post_hardware_snapshot(...) to sample and transmit hardware metrics in a single call.

4.3 Schema Drift Correction
During testing, we identified that power_draw_watts and carbon_intensity were present in the database model and output schema, but omitted from the input EventCreate schema. This caused incoming power draw values sent to /ingest/event to be ignored. We added both fields to EventCreate as optional attributes, allowing explicit power values and carbon intensity readings to persist.

4.4 Safe Database Migration (database/migrate.py)
• Implemented non-destructive run_migrations('ecogen.db').
• Queries PRAGMA table_info and applies individual ALTER TABLE ADD COLUMN statements only when missing.
• Preserves 100% of existing rows without table dropping or recreation.
• Triggered automatically on FastAPI application startup (@app.on_event('startup')).

5. Verification & Test Matrix

Check	Test Description	Expected Behavior	Result
Check 1	SQLite schema inspection (sqlite3 ecogen.db '.schema')	All 10 new columns in inference_events, 2 in hardware_snapshots.	**PASSED Verified**
Check 2	Full-payload ingestion via POST /ingest/event	HTTP 200; all 10 new fields (provider, pue, cost, MoE active params) populated.	**PASSED Verified**
Check 3	Minimal required fields only via POST /ingest/event	HTTP 200; defaults applied (pue=1.2, gpus=1, batch=1); nullable fields return null.	**PASSED Verified**
Check 4	Query events (GET /events?limit=5 and GET /events/1)	Pre-migration rows deserialize cleanly with nulls; new rows show populated fields.	**PASSED Verified**
Check 5	Unmodified decorator test (Gemini & Ollama @track_inference)	Function returns original response; last_event captures provider and tokens; auto-posted to DB.	**PASSED Verified**
Check 6	Hardware snapshot ingestion (POST /ingest/hardware)	Explicit and ?sample_idle=true baseline captures succeed and persist.	**PASSED Verified**

6. Hand-off to Layer 2 & Layer 3

With Layer 1 finalized and hardened, the data contract between layers is locked:
• **Layer 2 Readiness:** Can compute dynamic inference energy: $E_{inf} = (P_{active} - P_{idle}) \times runtime \times PUE$. Can amortize embodied impact using hardware_install_date, and apply Tier 2 FLOPs fallbacks using active_param_count_b.
• **Layer 3 Readiness:** Can segment carbon intensity and operational cost by provider, department, and use_case to recommend greener model architectures and batching schedules.
• **Repository Status:** Ready for staging/production commit with zero schema breaking changes.